FATIMA JINNAH WOMEN UNIVERSITY

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INTERNSHIP REPORT

Submitted in Partial Fulfilment of the Requirements for the Degree of
BS Computer Science

Organization

Attock Refinery Limited (ARL)

Web Development Project undertaken for Attock Sahara Foundation (ASF)

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Executive Summary

This report documents my internship experience as an IT Intern at Attock Refinery Limited (ARL), working within the Information Technology department under the supervision of Sir Kashif Ali. Over the course of six weeks, my work was split into two distinct phases: the first three weeks were spent working hands-on inside IBM Maximo Asset Management — covering assets, service requests, work orders, preventive maintenance, and procurement — and the final three weeks were spent designing, structuring, and debugging a web page for Attock Sahara Foundation (ASF) using WordPress and Elementor, both of which were entirely new tools to me at the start of that phase.

My work in the Maximo phase involved exploring the platform's Asset, Work Order Tracking, Preventive Maintenance, and Purchasing applications in enough depth to understand how ARL tracks equipment health, schedules maintenance, and manages procurement end-to-end. In the web development phase, I moved from having no prior exposure to WordPress or Elementor to independently restructuring a page's container hierarchy, fixing a horizontal button-alignment issue, and diagnosing a genuine browser-level file-handling defect that was forcing automatic downloads instead of in-browser document viewing.

My work was reviewed on a regular basis by my supervisor, which meant the tasks I completed carried a real quality bar rather than being purely academic exercises. This report covers an overview of the organization, a week-wise account of the duties I performed, a SWOT-based analysis of the organization from my vantage point as an intern, recommendations arising from that analysis, and a reflection on what the internship contributed to my technical and professional development.

Taken together, the two phases of this internship gave me a fairly rare combination of exposure: the discipline and structure of working inside a mature, enterprise-grade maintenance and asset-management system, alongside the independence of being handed unfamiliar tools and a real deliverable with no one to lean on but my own research and testing. I have tried to reflect that balance honestly throughout this report, rather than presenting either phase as more polished than it actually was at the time.

Organization Overview

Attock Refinery Limited (ARL) was originally established as a Private Limited Company in November 1978, taking over the oil refining and petroleum supply operations previously run by the Attock Oil Company Limited (AOC). It became a Public Limited Company in June 1979, and its shares are presently traded on all three stock exchanges in Pakistan; the company is also a registered participant of the Central Depository Company of Pakistan (CDC). Its paid-up capital, originally around Rs 80 million at inception, has since grown to approximately Rs 852.93 million.

ARL holds the distinction of being Pakistan's first oil refining company, with operations tracing back to 1922, when two small refining units began processing oil at Morgah following the earlier discovery of oil at Khaur in 1915. Over more than ninety years, the refinery has been repeatedly expanded and modernized — from the addition of new distillation, thermal cracking, and solvent extraction units in the late 1930s and 1940s following the Dhulian discovery, to the commissioning of larger-capacity distillation units in 1981 after further reserves were confirmed at Meyal and Toot, through to the completion of a Captive Power Plant, a Heavy Crude Unit, and a Catalytic Reformer around 1999–2000.

Refinery Products

ARL produces a wide range of fuel and non-fuel petroleum products, including Liquefied Petroleum Gas (LPG), unleaded Petroleum Solvent Grade (PMG), naphtha, motor gasoline, JP-1, JP-8, kerosene oil, High Speed Diesel (HSD), Light Diesel Oil (LDO), Furnace Oil (FO), Low Sulfur Fuel Oil (LSFO), and paving-grade asphalt. It also exports several of these products — including naphtha, kerosene oil, HSD, and JP-8 — which reflects its standing as an established, quality-conscious refiner within the regional petroleum industry.

Competitive Landscape

ARL operates within Pakistan's small group of established oil refining companies, alongside players such as Pakistan Refinery Limited, National Refinery Limited, and PARCO. What sets ARL apart, from what I observed during my time there, is its long operating history and the maturity of its enterprise systems — the tight integration between Maximo, Oracle Financials, and HRMS gives it a level of operational visibility that a newer or less established refiner would likely need years to build.

Organizational Design

ARL follows a functional departmentalization model, with distinct departments responsible for plant operations, maintenance, procurement, finance, human resources, and information technology. The IT department, where I was based, functions as a support and enablement unit — maintaining and administering Maximo, Oracle Financials, and HRMS, and coordinating with other departments to translate their operational needs into system configurations.

- Chain of Command: I reported directly to Sir Kashif Ali, who assigned my weekly tasks, reviewed my progress, and provided guidance whenever I got stuck — particularly during the earlier Maximo-focused weeks.
- Span of Control: The department's structure allowed for close, hands-on supervision rather than infrequent check-ins, which I found especially useful while I was still building familiarity with Maximo's modules.
- Work Specialization: My tasks shifted meaningfully partway through the internship — from enterprise-system exploration within Maximo to independent, deliverable-driven web development for Attock Sahara Foundation — giving me exposure across two fairly different technical domains rather than a single narrow role.
- Departmental Interaction: Although I was formally based within the IT department, my work regularly touched the boundaries of maintenance, procurement, and finance — since Maximo's Work Order and Purchasing modules exist specifically to coordinate across those functions — which gave me a clearer sense of how a large industrial organization's departments depend on one another rather than operating in isolation.

Looking back, this organizational design shaped the internship experience in a very direct way: the flat, closely-supervised structure of the IT department meant I was never left to guess what "good" looked like, while the cross-departmental nature of the systems I worked with meant I ended up understanding far more of ARL's operations than my job title alone would suggest.

Week-wise Description of Duties

Position Title: IT Intern — Enterprise Systems & Web Development

Department: Information Technology Department, Attock Refinery Limited (ARL). I spent the first three weeks of the internship working within ARL's core IT function on the IBM Maximo Asset Management system, and the final three weeks on an assigned web development project for Attock Sahara Foundation (ASF), an affiliated community initiative.

The table below summarizes my internship on a week-wise basis, covering the period from August 13, 2026 to September 23, 2026.

Week	Focus Area	Tasks Completed
Week 1 *Aug 13 – Aug 19, 2026*	**Orientation & Organizational Familiarization**	I began the internship with a structured orientation covering ARL's history, its position within Pakistan's petroleum refining sector, and the organizational structure of the departments I would be interacting with. My supervisor, Sir Kashif Ali, walked me through the IT department's role as a support and enablement function and gave me a high-level tour of the three core enterprise systems in use across the organization — IBM Maximo, Oracle Financials, and HRMS — along with how invoice, vendor, and employee data flows between them. I also reviewed the organization's IT policies and data-handling protocols that would apply to my work for the remainder of the internship.
Week 2 *Aug 20 – Aug 26, 2026*	**IBM Maximo — Assets & Service Requests**	I started working hands-on inside IBM Maximo, focusing first on the Assets application. I went through asset records, meters, locations, and specifications to understand how physical plant equipment is represented digitally, and I practiced creating and viewing Service Requests — populating fields such as Reported By, Summary, Asset, Location, and Priority, then tracking status changes as a request moved through the workflow. I also studied the Failure Codes application and the Problem–Cause–Remedy hierarchy Maximo uses to categorize equipment failures, which gave me a clearer picture of how maintenance issues are logged and diagnosed.
Week 3 *Aug 27 – Sep 02, 2026*	**Work Orders, Preventive Maintenance & Procurement**	This week I moved into the operational core of Maximo. I traced the full lifecycle of a work order through the Work Order Tracking application — from creation and assignment through to execution and closure — and reviewed the Preventive Maintenance module, distinguishing between

Week	Focus Area	Tasks Completed
		Master PM templates and the site-specific PM records that actually generate scheduled work orders based on time or meter frequency. I also worked through the Purchasing workflow end-to-end, from Purchase Requisition to Purchase Order, Receiving, and Invoices, and reviewed the Item Master application within Inventory to see how storeroom stock is defined and tracked. By the end of the week I had a working, module-level understanding of how Maximo ties maintenance planning to procurement and inventory.
Week 4 *Sep 03 – Sep 09, 2026*	**Transition to Web Development — Learning WordPress & Elementor**	I was assigned a new project this week: designing and structuring web pages for Attock Sahara Foundation (ASF), a community initiative affiliated with ARL. WordPress and Elementor were both completely new to me, so I spent this week teaching myself the basics — the WordPress dashboard, page and post structures, the Media Library, and Elementor's container-based layout system and widget library. I reviewed the organization's existing "Programs" page as a reference template and began drafting the outer structure of a new "Accounts" page that I would need to build to match it.
Week 5 *Sep 10 – Sep 16, 2026*	**Layout Structuring & Container Troubleshooting**	I ran into two layout problems while building the "Accounts" page and worked through both independently. First, stacking the section heading directly above the action buttons without a proper container hierarchy was causing elements to misalign, so I restructured the page with an outer container set to a vertical (Column) direction, which correctly anchored the header at the top with a dedicated inner container beneath it for the buttons. Second, my "View" and "Download" buttons were stacking vertically instead of sitting side by side, which I fixed by setting that inner container to a horizontal (Row) direction. I tested the layout across a few different screen sizes to confirm it held up responsively.
Week 6 *Sep 17 – Sep 23, 2026*	**Link Behavior Debugging & Final Delivery**	In my final week, I diagnosed why both the "View" and "Download" buttons were forcing automatic downloads instead of opening documents in a new tab. I traced this to two causes: the linked files were .zip archives, which

Week	Focus Area	Tasks Completed
		browsers force-download regardless of target-window settings, and a set of orphaned hash symbols (#) left over from earlier edits, which were causing redirect loops back to the /accounts/ page. I cleaned up the button attributes, made sure "Open in new window" was enabled, and set up a rule to convert all viewable financial documents from .zip to .pdf in the Media Library going forward. I then ran a final round of cross-browser testing before handing off the completed, working "Accounts" page.

Table 1: Week-wise Description of Duties

Key Deliverables

- A working, hands-on understanding of IBM Maximo's Asset, Service Request, Work Order, Preventive Maintenance, and Purchasing applications, built through direct exploration rather than documentation alone.
- A completed, tested "Accounts" page for the Attock Sahara Foundation website, matching the layout and behavior of the organization's existing "Programs" page.
- A resolved browser-level file-handling defect, along with a documented rule (converting .zip files to .pdf) to prevent the same issue from recurring on future uploads.

Across both phases, the pattern that stood out to me most was how much of the work involved reading and understanding an existing system — whether that was Maximo's module structure or an unfamiliar page's container hierarchy — before I was able to contribute anything meaningfully. That skill, more than any single tool, is what I feel I leave this internship with.

Organizational Analysis

The following SWOT analysis reflects my own observations as an intern working within ARL's IT department, rather than internal organizational data. I have tried to ground each point in something I directly encountered during the six weeks, rather than generic statements that could apply to any large organization.

Because my internship spanned two very different kinds of work — enterprise system exploration within Maximo, and independent web development for an affiliated community platform — this analysis draws on both halves of the experience. The strengths and weaknesses I noticed in the Maximo phase speak to how well-established ARL's core systems are; the opportunities and threats I noticed in the web development phase speak more to the smaller, less formalized digital properties sitting just outside that core.

S — Strengths	**W — Weaknesses**
ARL's ninety-plus-year operating history and its position as Pakistan's first oil refining company give it strong institutional credibility. During my time in the IT department I saw firsthand how tightly its three core systems — Maximo, Oracle Financials, and HRMS — are integrated, letting maintenance, procurement, and financial data flow between departments without manual re-entry, which is not something every organization of this size manages well.	Several of the Maximo applications I worked with (particularly the Oil & Gas-specific ones) carry a steep learning curve with limited onboarding material, which meant I relied heavily on direct supervisor guidance rather than documentation. I also noticed that ARL's affiliated community platforms, such as the ASF website, had not been built to any documented layout or file-handling standard, which is part of why the issues I fixed in Weeks 5 and 6 existed in the first place.
O — Opportunities	**T — Threats**
The same integration ARL has already achieved across Maximo, Oracle Financials, and HRMS could be extended to its smaller, affiliated digital properties like the ASF website, bringing them up to a similar standard of consistency and reliability. There also seems to be room to formalize a small internal web-development capability, potentially staffed by interns, given how quickly I was able to pick up WordPress and Elementor and deliver a working page.	Increasingly strict browser security policies around file handling and downloads (the exact issue I debugged in Week 6) are likely to keep surfacing on any legacy or undocumented web pages until a consistent file-format standard is enforced. I'd also note that specialized knowledge of Maximo's Oil & Gas-specific modules appeared concentrated in a small number of staff, which could be a continuity risk.

Table 2: SWOT Analysis Summary

Taken together, these observations point to a consistent pattern: ARL's core enterprise systems are mature, well-integrated, and reliable, but that same standard has not yet been extended to smaller, affiliated digital initiatives like the ASF website. The recommendations in the following section are built directly around closing that gap.

It is also worth noting that this gap is not unusual for an organization of ARL's size and history. Large, established companies frequently invest heavily in their core operational systems — in ARL's case, the systems that directly touch plant maintenance, procurement, and finance — while smaller, affiliated, or community-facing platforms receive comparatively less structured attention simply because they sit outside the primary business function. What stood out to me was less that this gap existed, and more how clearly it showed up in practice: the same kind of undocumented convention that would never survive in Maximo (which has years of configuration, documentation, and dedicated administrators behind it) was exactly what caused the Week 5 and Week 6 issues on the ASF website.

From my perspective as an intern moving between both environments in the same six weeks, this contrast was one of the more instructive parts of the internship: it gave me a concrete, side-by-side comparison of what a well-governed system looks like versus one that has grown organically without the same oversight, and it shaped several of the recommendations that follow.

Recommendations

Based on the observations summarized in the SWOT analysis above, I would offer the following recommendations to help ARL build on its existing strengths and address the weaknesses and threats I encountered:

- Extend the same standard of system integration and documentation that governs Maximo, Oracle Financials, and HRMS to smaller, affiliated digital properties such as the ASF website, so they are held to a consistent technical standard rather than being maintained ad hoc.
- Establish a documented file-format and layout standard for community-facing WordPress pages — for example, mandating .pdf over .zip for viewable documents — so that issues like the one I debugged in Week 6 are prevented rather than discovered after the fact.
- Build a short internal onboarding guide for Maximo's more specialized (particularly Oil & Gas-specific) applications, which would have shortened my own ramp-up time considerably during Weeks 2 and 3.
- Continue assigning interns real, deliverable-driven work reviewed by a supervisor rather than simulated exercises — this was, by a wide margin, the most valuable part of my own experience and is worth preserving as the internship program grows.
- Consider formalizing a small internal web-development capability, potentially staffed by trained interns, given how quickly I was able to learn WordPress and Elementor and deliver a working, tested page from a standing start.
- Introduce a brief pre-launch checklist for any new or edited web page (checking file formats, link targets, and cross-browser behavior) so that issues like the one I debugged in Week 6 are caught before a page goes live rather than after.
- Where practical, pair interns working on affiliated platforms like the ASF website with a technical point of contact from the core IT team, so that lessons learned on the main systems (such as the value of documented conventions) carry over to the smaller properties as well.

Learning Outcomes

Application of Theoretical Knowledge

This internship gave me a direct, practical extension of concepts I had only encountered in coursework — particularly around database concepts, system architecture, and front-end web technologies. Understanding how Maximo's modules relate to one another, and how data moves between Maximo, Oracle Financials, and HRMS, drew directly on what I had studied about relational systems and system integration. Similarly, debugging Elementor's container hierarchy and the browser's file-download behavior required a level of precision and independent reasoning that coursework exercises rarely demand, since there was no answer key — only a working page or a broken one.

Key Achievements

- Built a working, module-level understanding of IBM Maximo across its Asset, Work Order, Preventive Maintenance, and Procurement functions within three weeks.
- Learned WordPress and Elementor from zero prior exposure and used them to independently design, structure, and debug a fully functional web page for a real organizational stakeholder.
- Diagnosed and resolved a genuine browser-level file-handling defect (forced downloads caused by .zip files and orphaned hash-symbol redirects) rather than a simulated or textbook bug.
- Practiced professional workflow habits, including taking direction from a supervisor, iterating on feedback, and testing my own work across scenarios before considering it complete.

Responsibilities Liked and Disliked

I found the Week 3 work on Preventive Maintenance particularly engaging, since it showed me how a well-configured system can meaningfully reduce downtime and operational risk in a physical, industrial setting — something that felt much more concrete than anything I had studied in the classroom. The Week 6 debugging work was similarly rewarding, since it required forming and testing a hypothesis about why the buttons were misbehaving rather than simply following a set of instructions. What I found most challenging, at least initially, was Week 4 — being handed two completely unfamiliar tools (WordPress and Elementor) with a real deliverable expected meant a fair amount of trial and error before I felt productive, but it ended up being one of the most useful weeks of the internship.

Relevance to Career Interests

This internship is directly relevant to my interest in enterprise systems and applied software development. My exposure to IBM Maximo gave me a credible, resume-relevant foundation in enterprise asset management — a domain with real demand across industrial and utilities sectors — while the WordPress and Elementor work strengthened my practical front-end

development and debugging skills, which I expect to draw on directly in future technical roles and interviews.

More broadly, the internship confirmed something about how I want to work: I found I was consistently more motivated by problems that had a clear, verifiable outcome — a service request that either got logged correctly or didn't, a button that either opened a document or force-downloaded it — than by open-ended theoretical exercises. That preference is something I plan to keep in mind as I start narrowing down the kind of roles I apply for after graduation.

Skills Gained

- Practical, applied familiarity with enterprise asset management concepts (assets, work orders, preventive maintenance, procurement) as implemented in a real, industry-grade system.
- Front-end structuring and debugging skills in WordPress and Elementor, including container-based layout logic and responsive design troubleshooting.
- A working understanding of browser file-handling behavior and how file formats (.zip vs. .pdf) and link attributes affect in-browser rendering.
- Stronger habits around independent diagnosis — forming a hypothesis about why something is broken, testing it, and narrowing down the actual root cause rather than guessing.

Additional Comments

I would note, for future interns entering a similarly structured internship, that the shift from a well-established enterprise platform (Maximo) to a self-taught, hands-on web development assignment was an unexpectedly effective way to learn — the earlier weeks gave me close mentorship while I was still building confidence, and the later weeks gave me the independence to actually solve a real problem on my own. I would recommend approaching the second half of any similar internship with the expectation that you will need to teach yourself the tools before you can be expected to use them well.

Overall, I am leaving this internship with a much clearer picture of what enterprise IT work actually looks like day to day — less about writing code from scratch, and more about understanding an existing system well enough to extend it, fix it, or explain it to someone else. That is a lesson I don't think I could have gotten from coursework alone, and I am grateful to Sir Kashif Ali and the rest of the IT team at ARL for the opportunity.

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Annexure

The following supporting documents are attached in support of this report:

- Copy of the Internship Certificate on the organization's official letterhead.
- Original Evaluation Performa, duly signed and stamped by the internship supervisor/organization.